

2.2 Projects Developed During the Internship

2.2.1 Smart Industrial Safety and Employee Monitoring System

➤ Introduction

The Smart Industrial Safety and Employee Monitoring System is an IoT-based real-time monitoring solution developed using Raspberry Pi 4, environmental sensors, RFID technology, and the Ubidots cloud platform. The system is designed to enhance industrial safety by continuously monitoring environmental conditions such as gas leakage and fire hazards, while also implementing employee identification and access control.

Industrial environments are exposed to risks like gas leaks, fire accidents, overheating, and unauthorized access. Manual monitoring can cause delays in detecting such hazards. To overcome these challenges, this system provides automated monitoring, immediate local alerts, and remote cloud-based supervision. The project integrates embedded hardware and IoT technology to create a reliable and efficient safety solution.

➤ Components Used

Hardware Components:

- Raspberry Pi 4 – Central processing and control unit
- MQ Series Gas Sensor – Detects harmful gases and smoke
- Flame Sensor – Detects fire or flame presence
- RFID Module – Used for employee identification
- RFID Tags/Cards – Assigned to employees for access verification
- Relay Module – Controls electrical devices during emergencies
- LED Indicators – Visual warning signals
- Buzzer – Audio alert system
- Breadboard – Circuit prototyping
- Jumper Wires – Electrical connections

Software & Platform Used:

- Python Programming Language – For system logic and sensor data processing
- Ubidots IoT Platform – For real-time cloud monitoring and data visualization

➤ Block Diagram

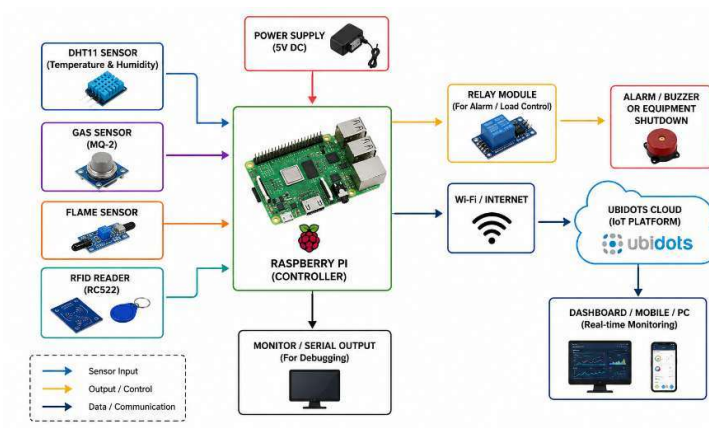


Figure 2.1: Block Diagram of Smart Industrial Safety and Employee Monitoring System

➤ Hardware Implementation

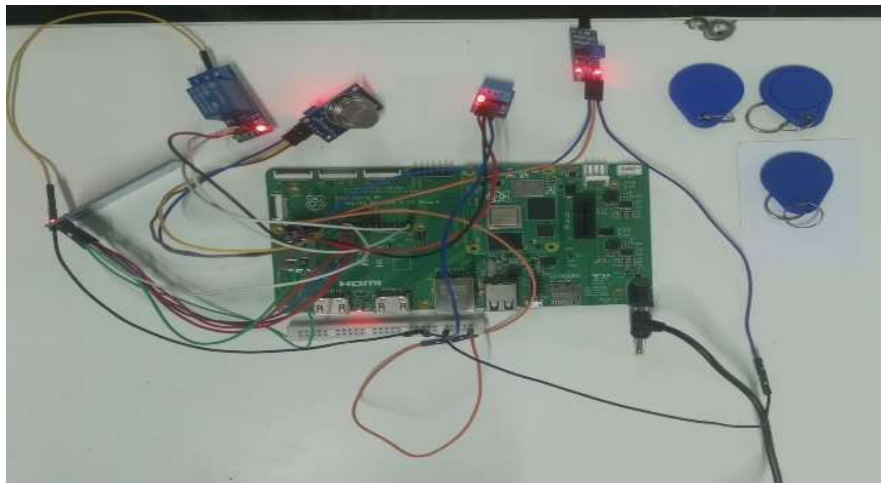


Figure 2.2: Hardware Setup of Industrial Safety System

➤ Working of the System

The system operates continuously to monitor industrial safety conditions in real time.

1. The gas sensor continuously checks air quality and detects harmful gases or smoke.

2. The flame sensor detects fire by sensing infrared radiation.
3. The RFID module scans employee cards to verify authorized access.
4. All sensor data is sent to the Raspberry Pi 4, which processes the information using Python programming.
5. If hazardous conditions are detected:
 - The LEDs turn ON.
 - The buzzer activates to generate an alarm.
 - The relay module can disconnect connected equipment for safety.
6. Sensor data is transmitted to the Ubidots cloud platform through internet connectivity.
7. The industrial conditions can be monitored remotely using the cloud dashboard in real time.

This system ensures automated safety monitoring, instant alert generation, and remote accessibility.

➤ **Results Obtained**

The system was successfully implemented and tested. The following results were observed:

- The gas sensor successfully detected smoke and harmful gases.
- The flame sensor accurately detected fire conditions.
- The RFID system worked properly for employee identification and access verification.
- LEDs and buzzer provided immediate warning alerts during hazardous situations.
- Sensor data was successfully transmitted to the Ubidots IoT dashboard.
- Real-time monitoring was achieved through cloud connectivity.
- The Raspberry Pi effectively controlled the entire system.

➤ **Table: Sensor Readings Under Normal and Abnormal Conditions**

Parameter	Normal Condition	Abnormal Condition (Gas/Fire Detected)
Temperature	25–30 °C	Above 40 °C
Humidity	40–60 %	Below 30 % (or sudden change)
Gas Indicator	OFF (0)	ON (1)
Flame Indicator	OFF (0)	ON (1)
LED Indicator	OFF	ON
Relay Status	ON (System Running)	OFF (Safety Mode Activated)

Table 1.1 Sensor Readings

➤ **Ubidot's Dashboard Realtime readings**

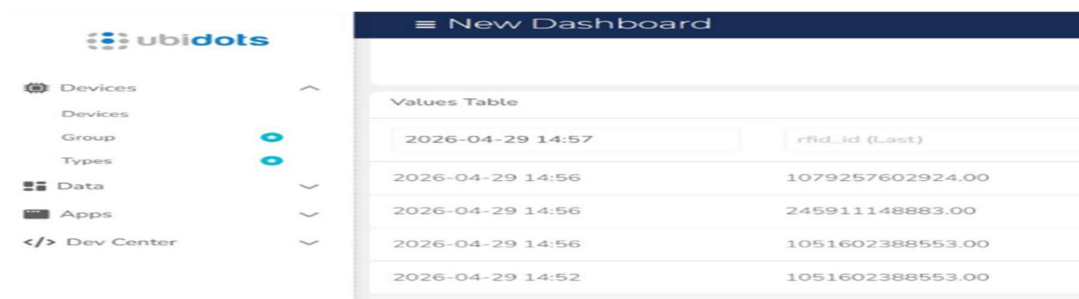


Figure 2.3: RFID Card Reading Displaying Employee ID Number for Authentication



Figure 2.4: Ubidots Dashboard Showing Sensor Monitoring

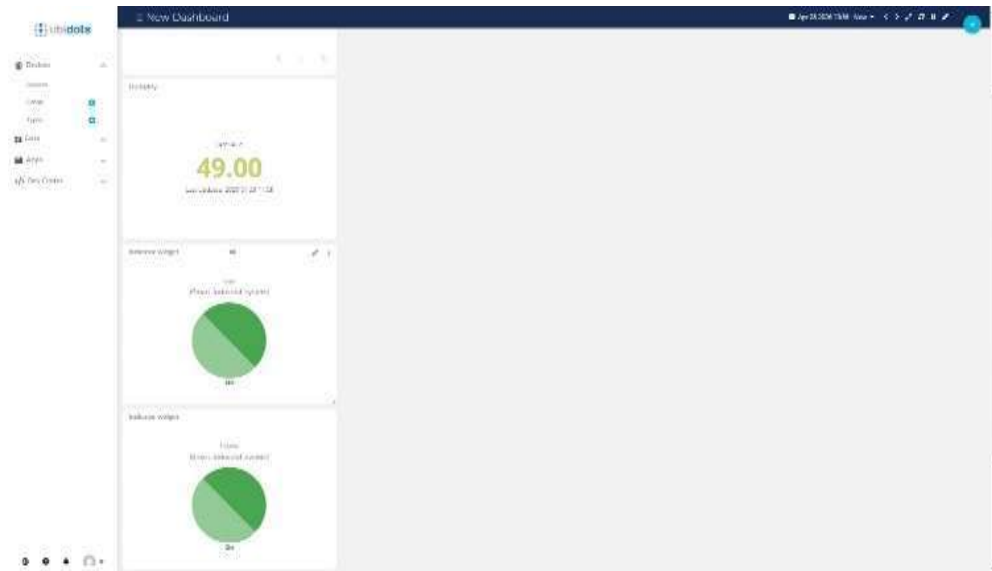


Figure 2.4: Ubidots Dashboard Showing Sensor Monitoring

```

Sensor Data
-----
Temperature : 27.0 °C
Humidity    : 46.0 %
Gas         : ON
Flame       : ON
Response    : 200
-----
Scan RFID Card...
Card ID : 1079257602924
Name    : shikha
Sending to Ubidots: {'rfid_id': {'value': 1079257602924, 'context': {'name': 'shikha'}}, 'temperature': 0, 'humidity': 0, 'gas': 1, 'flame': 1}

Sensor Data
-----
Temperature : 0 °C
Humidity    : 0 %
Gas         : ON
Flame       : ON
Response    : 200
-----
Scan RFID Card...
Card ID : 1041392141900
Name    : russian embassy
Sending to Ubidots: {'rfid_id': {'value': 1041392141900, 'context': {'name': 'russian embassy'}}, 'temperature': 27.0, 'humidity': 46.0, 'gas': 1, 'flame': 1}

Sensor Data
-----
Temperature : 27.0 °C
Humidity    : 46.0 %
Gas         : ON
Flame       : ON
Response    : 200
-----
    
```

Figure 2.5: Real-Time Sensor Output Display